

MineTrac AI — Dashboard & Voice Notes Master Guide (SIH26)

Complete Technical Breakdown, Minute Button Details, 3-Tier ML Pipeline & Word-for-Word Hinglish Voice Scripts

■ Purpose of this Master Guide:

Yeh document aapki poori team aur dosto ke liye ek comprehensive reference runbook hai. Isme MineTrac Dashboard ke **har minute button**, **card**, **underground schematic**, **Leaflet map**, **3-Tier ML model prediction logic**, **4 longwall scenarios**, **word-for-word Hinglish voice scripts**, aur **ESP32 hardware deployment steps** ko deep detail me explain kiya gaya hai.

1. Dashboard Header & Minute Button Breakdown

1. Logo & Team Branding:

- **MineTrac Logo:** MineTrac BY AKSSAI ProjExcel | SIH26 Coalition. AI-driven underground mining risk platform branding.

2. Header Control Buttons (Minute Detail):

- ■ Reports Button: Opens a modal generating downloadable PDF & CSV compliance audit logs for Directorate General of Mines Safety (DGMS).
- ■ All Mesh Nodes Button: Opens a grid modal displaying individual hardware status, RSSI radio strength, and battery voltage of Nodes N1–N5.
- ■■ Safety Rules Button: Displays DGMS safety thresholds (Max allowable tilt = 0.5° , Max strain = $15\ \mu\epsilon$).
- ■■ Master Config Button: Adjusts Gateway polling frequency (1s to 5min) and SMS notification phone numbers.
- ■ Audio Siren: MUTED / ACTIVE Button: Controls Web Audio API synthesizer. When active during Critical Subsidence (Phase 3), fires a dual-tone emergency siren.
- ■ User Badge: Justin Humphrey: Displays current logged-in role (Safety Director, Jharia Coalfield Panel 4).

3. Secondary Navigation Tabs:

- Dashboard: Primary operational cockpit displaying live KPIs, schematic, GIS map, and early warning gauge.
- ML Test Bench (3-Tier ML): Interactive ML playground with 8 sensor sliders for testing Edge, Gateway, and Cloud model inferences live.
- 90-Day Simulation & Scenarios: Time-series playback of the 4 longwall extraction phases (50 windows).
- Hardware & Flashing Guide: Hardware pinout diagrams and Arduino C source code.

2. Top KPI Summary Cards & Test Presets Bar

Top KPI Summary Cards:

- **Active Mesh Nodes (5/5 Online):** Shows 100% 802.15.4 mesh radio connectivity. If a node loses power/signal for >30s, status updates to *Offline/Stale*.
- **Max Surface Tilt (0.02°):** Live ground inclination angle calculated from MPU6050 accelerometer. Baseline = 0.02° .
- **Max Strain Stress Delta ($0.5\ \mu\epsilon$):** Microstrain delta from digital strain gauge. Baseline = $0.5\ \mu\epsilon$.
- **Subsidence Risk % (2.4% SAFE_STABLE / 12.0% BLAST_FILTERED):** AI-calculated collapse probability.

Test Presets Bar (1-Click Scenario Simulation):

- Case 1: Normal Operation: Safe baseline (Tilt 0.02° , Strain $0.5\ \mu\epsilon$, Vib 0.025g, Risk 2.4%).
- Case 2: Pre-Subsidence Creep: Monotonic tilt ramp ($0.05^{\circ} \rightarrow 0.35^{\circ}$), strain bump ($8.5\ \mu\epsilon$), Risk 45.0%.
- Case 3: Active Subsidence: Rapid collapse (Tilt 1.45° , Strain $32\ \mu\epsilon$, Breakwire broken), Siren fires, Risk 94.2%.
- Case 4: Explosive Blasting: High vibration spike (1.45g) with ZERO tilt/strain shift. Blast Filtered!
- Case 5: Breakwire Trip: Direct physical fissure rupture detection.

3. Underground Longwall Schematic & GIS Satellite Map

Underground Longwall Mine Cutaway Schematic:

- **Surface Layer:** Coal Preparation Plant, Raw-Coal Silo, Elevator Shaft Head, and 5 Surface Sensor Pins (Slave N1 to N5) spaced over 1500m × 200m grid.
- **Strata Layer:** Overburden Sandstone & Shale Strata. Transmits underground roof collapse strain upwards.
- **Underground Layer:** Elevator Shaft, Haulage Slope, **GOB ZONE (Collapsed Roof Void)**, Hydraulic Roof Support Shearer, Continuous Miner, and Coal Pillars.
- **Dynamic Visualization:** As ground deformation occurs, GOB zone borders and surface node pins dynamically transition from Green → Amber → Red.

GIS Satellite Map (Leaflet.js + Esri World Imagery):

- Pinpoints nodes N1–N5 on actual Jharia Coalfield GPS coordinates.
- Node N3 represents the center max-stress zone.
- Clicking any node marker opens a live popup with pitch, roll, tilt, vibration RMS, battery voltage, and TinyML score.
- **Color Legend:** ■ Normal | ■ Pre-Subsidence | ■ Active Alert | ■ Blast Filtered

4. Three-Tier ML Model Architecture & Prediction Logic

ML Stage & Layer	Model Type & Framework	Execution Speed & Function
Stage 1: Edge TinyML (ESP32-S3 Node)	Isolation Forest Compiled C Header (isolation_forest_model.h)	101.3 μs execution. Evaluates 8 scaled feature vectors. Normalized score $s = -2^{(-avg_depth / c(n))}$. Flags score < -0.6127.
Stage 2: Gateway Correlator (Master Gateway)	Spatial-Temporal Rules Engine Adjacent Node Correlator	Evaluates spatial multi-node consensus. Detects high vibration with ZERO tilt/strain → OPERATIONAL BLAST (FILTERED), suppressing siren.
Stage 3: Cloud LSTM (Render Cloud)	Multi-Output LSTM TensorFlow Keras (24hr window)	Predicts Subsidence Probability %, Max Displacement (mm), and Severity Class (LOW / MEDIUM / HIGH).

5. Word-for-Word Hinglish Voice Note Scripts (Voice 1 to 7)

■ ■ Voice Note 1: Top Navigation & Executive Introduction

"Hello everyone! Welcome to MineTrac AI. Yeh humara low-cost, real-time mine subsidence detection, prediction, aur early warning platform hai jo Smart India Hackathon (SIH26) ke problem statement ke liye develop kiya gaya hai. Top bar par aap mine location dropdown dekh sakte hain jo abhi Jharia Coalfield Longwall Panel 4 ko monitor kar raha hai. Yahan active navigation tabs hain—Dashboard, 3-Tier ML Test Bench, 90-Day Simulation Timeline, aur Hardware Flashing Guide. Right side par audio siren control abhi Muted state me hai aur safety director badge gateway connectivity verify karta hai."

■ ■ Voice Note 2: Top KPI Metric Cards & Test Preset Buttons

"Top KPI cards par focus karte hain: Pehla card hai Active Mesh Nodes, jo dikhata hai ki hamare 5 me se 5 nodes online hain aur 100% radio signal ke saath connected hain. Doosra card hai Max Surface Tilt jo 0.02 degrees par hai—yeh hamara normal safe baseline hai. Teesra card hai Max Strain Stress Delta jo 0.5 microstrain par normal tensile strain dikhata hai. Chautha card hai Subsidence Risk Percentage jo 2.4% par Safe Stable status dikha raha hai. Directly neeche 5 quick test preset buttons hain jisse hum Normal, Pre-Subsidence, Active Collapse, aur Dynamite Blasting scenarios ko 1-click me test kar sakte hain."

■ ■ Voice Note 3: Underground Mine Cutaway Schematic

"Yeh middle section me hamara anatomical underground longwall mine cross-section visualizer hai. Top surface par hamare preparation plant, coal silo, elevator shaft head, aur 1500m × 200m surface grid par Slave N1 se Slave N5 tak ke 5 sensor pins mounted hain. Neeche overburden sandstone aur shale strata layers hain jo underground ground deformation ko surface tak transmit karti hain. Sabse neeche underground area me GOB Zone—yaani collapsed roof void—hydraulic roof support shearer, aur coal pillars dikhaye gaye hain. Jaise hi underground caving start hoti hai, surface pins aur GOB zone border ka color real-time dynamic update hota hai."

■ ■ Voice Note 4: GIS Satellite Map & Node Mesh Overlay

"Neeche left side par hamara Surface Panel GIS Map hai jo High-Definition Esri World Imagery Satellite Tiles use karta hai. Yeh map Jharia Coalfield ke actual GPS coordinates par surface sensor nodes ko pinpoint karta hai. Node N3 center max-stress zone par positioned hai. Marker par click karne se node ka real-time tilt, strain, vibration, battery, aur TinyML isolation forest status popup me khul jaata hai. Ground risk ke aadhar par color dots green, yellow, red, ya purple me transform ho jaate hain."

■ ■ Voice Note 5: Gateway Early Warning Status & Blast Filter

"Bottom right side par hamara Master Gateway Early Warning Status panel hai jo 0.0 se 1.0 ke scale par live subsidence risk score dikhata hai. Jab underground dynamite blasting hoti hai, toh traditional sensors false alarm trigger kar dete hain aur mining ko faltu me rokna padta hai. Lekin hamara Stage 2 Gateway Spatial Correlator detect karta hai ki high vibration ke saath TILT 0 hai. Ise yeh OPERATIONAL BLAST (FILTERED) classify karke physical siren ko suppress kar deta hai, jisse false alarm rukta hai aur mining bina kisi rukaawat ke chalti rehti hai!"

■ ■ Voice Note 6: Interactive 3-Tier ML Test Bench & Sliders

"Ab hum ML Test Bench tab par hain. Yahan Node N3 ke 8 physical sensor sliders—Tilt Mean, Tilt Rate, Strain Delta, Vibration RMS, Peak, Frequency, aur Crack Breakwire—ko live manipulate kar sakte hain. Right side par hamara 3-Tier ML Model inference result dikhata hai:

1. Stage 1 Edge TinyML (ESP32-S3) Isolation Forest C model ko 86.4 microseconds me execute karta hai.
2. Stage 2 Gateway Correlator adjacent nodes ka spatial correlation aur blast suppression evaluate karta hai.
3. Stage 3 Cloud Multi-Output LSTM (TensorFlow Keras) 24-hour sequence window se surface displacement millimeters me aur severity class predict karta hai."

■ ■ Voice Note 7: 90-Day Indian Longwall Mining Timeline (Phase 1 - 4)

"Finally, hamare 90-Day Simulation tab me hum Longwall Extraction ke pure 4 geomechanical phases ko 50 time-series windows me play back karte hain:

- Phase 1 (Normal, Days 0-60): Flat baseline ground (Tilt 0.02° , Risk 2.4%).
- Phase 2 (Pre-Subsidence Creep, Days 60-75): Overburden micro-cracking, steady tilt ramp ($0.05^\circ \rightarrow 0.37^\circ$), rising strain ($14.5 \mu\epsilon$), triggering early warning at 45% risk.
- Phase 3 (Active Collapse, Days 75-85): Main roof falls into GOB void, rapid tilt acceleration ($>1.5^\circ$), strain spike ($58 \mu\epsilon$), breakwire flip, triggering 94.2% risk & siren alert! ■
- Phase 4 (Dynamite Blast Event): Explosive vibration spikes to 1.45g while tilt stays ZERO (0.02°), demonstrating blast false-alarm suppression."

6. Remote Deployment Guide (Ghar + On-Site Hardware Team)

■ AAP (Ghar Par - Remote Developer):

1. Github repository https://github.com/Monaal5/mine_subsidence_ml.git par code updated hai.
2. **Render.com** par New Web Service banakar GitHub connect karein. Docker select karke Deploy par click karein.
3. 2 minute me Render aapko public link dega: <https://mine-subsidence-ml-2rr5.onrender.com>.
4. Modifies RAM usage under 200MB & dynamic PORT binding fixes free-tier memory issues!

■ HARDWARE TEAM (On-Site Location):

1. **ESP32-S3 Sensor Nodes Flashing:** ESP32-S3 ko USB se laptop me connect karein. Arduino IDE me `esp32_firmware/slave_node.ino` aur `master_gateway.ino` kholein.
2. Flash Master Gateway first, get its MAC address from Serial log, paste into Slave Node `masterAddress[]` array, and flash Slave.
3. Turn on Phone Hotspot WiFi — both nodes connect automatically and stream live hardware telemetry to the cloud dashboard!